

Math 4441: Probability and Statistics

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Chapter 1: Introduction (Part 2 – Conditional Probability)

Lectures 04 – 05

Math 4441: Probability and Statistics

References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Conditional Probability:

- Probability models when partial information about the outcomes of the experiment changes
 - Sample space
 - Probability of events
- The partial information can be defined as an event – it is given that the event has occurred
- Then the probability models or probabilities of events are dependent on the occurrence of the event
- Such a probability is known as conditional probability because it is applicable conditioned on the fact that the given event has occurred
 - The probability model is known as conditional probability model
- If B is the given event (partial information), the conditional probability of the event A conditioned on the fact that event B has already occurred is denoted by $P[A|B]$

Example 1.15: Roll a six-sided fair die. Before you observe the outcome, someone tells that the outcome is an even number. After knowing that the outcome is an even number, what is the probability the outcome 2?

Procedure: Roll a 6-sided fair die

Observation: Number of dots on the side facing upward

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A \equiv \{\text{Event that the outcome is 2}\}$$

$$B = \{\text{Event that the outcome is an even number}\} = \{2, 4, 6\}$$

Given that B has occurred, the sample is changed. The modified sample space (i.e., conditional sample space is

$$S_c = S \cap B = \{2, 4, 6\}$$

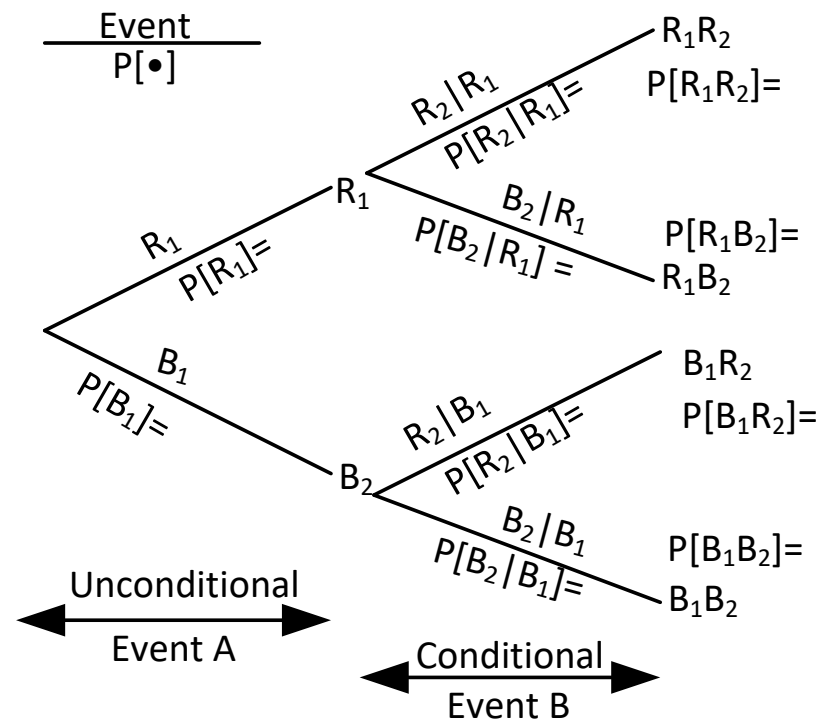
$$P[A|B] = \frac{1}{3}$$

Example 1.16: A box contains **8 red** balls and **5 blue** balls. Balls are mixed together. Pick two balls randomly one-after-another without replacement and observe their color. Find the following probabilities:

- a) The color of first ball is red
- b) The color of the second ball is red, if it is given that the first ball was red
- c) The colors of both the balls are red
- d) The color of the second ball is red
- e) The color of the first ball was red, if it is given that the second ball is red

$R_i \equiv$ Event that the i -th ball is red, $i = 1, 2$

$B_i \equiv$ Event that the i -th ball is blue, $i = 1, 2$



Conditional Probability

- Partial information is given, which changes the sample space or the probability of events.
- The probabilities are calculated in the modified sample space usually with less number of elements.
- Conditional probability is denoted by $P[A|B]$
- Probability of occurring event A , given that event B has already occurred
- Probability of A , given B

Conditional Probability is defined by

- Proportion of the outcomes in B represented by the joint event $A \cap B$
- If the outcomes are equally likely, ratio of the # of elements in $A \cap B$ to that in B
- In general, the ratio of $P[A \cap B]$ to $P[B]$, $P[B] > 0$ quantifies $P[A|B]$

$$P[A|B] =$$

Example: Roll a six-sided fair die and observe the number of dots on the top side

$$S = \{1, 2, \dots, 6\}$$

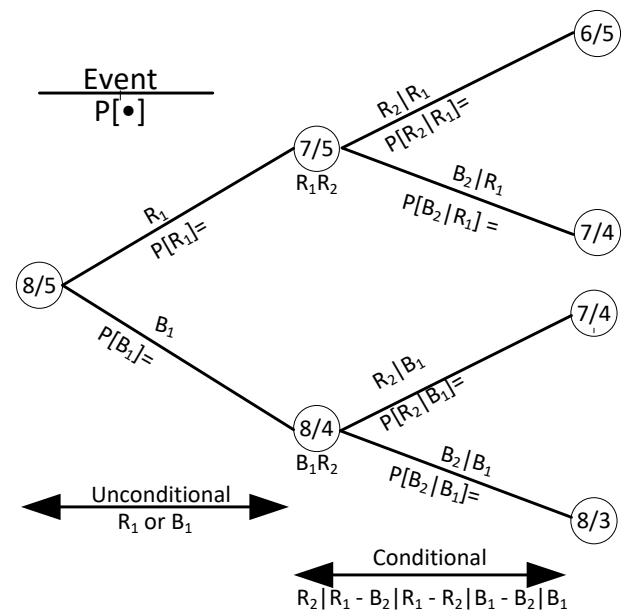
$$P[\{1\}] = \dots = P[\{6\}] = 1/6$$

$$B \equiv \{\text{event that the outcome is an even number}\}$$

A denote the event that outcome is 2

$$S|B = \{2, 4, 6\}$$

Example 1.17: A box contains **8 red** balls and **5 blue** balls. Balls are mixed together. Pick two balls randomly one-after-another without replacement and observe their color. R_i and B_i denote the events that the i -th ball is red and blue, respectively. Find $P[R_2|R_1]$.



When R_1 is given (color of the 1st ball is red)

$$S_c =$$

$$P[R_2|R_1] =$$

$$p[B_2|R_1] =$$

When B_1 is given (color of the 1st ball is red)

$$S_c =$$

$$P[R_2|B_1] =$$

$$p[B_2|B_1] =$$

Probability of Axioms (for Conditional Probability)

Axiom 1 (Non-negativity):

$$P[A] \geq 0$$

Axiom 2 (Additivity): $A_1 \cap A_2 = \phi$

$$P[A_1 \cup A_2] = P[A_1] + P[A_2]$$

Axiom 3 (Normalization):

$$P[S] = P[\cup_{i=1}^n \omega_i] = \sum_{i=1}^n P[\omega_i] = 1$$

Example:

$$S = \{1, 2, \dots, 6\}$$

$$P[\{1\}] = \dots = P[\{6\}] = 1/6$$

$B \equiv$ event that the outcome is an even number

A_2, A_4, A_6 denote the events that outcome is 2, 4 and 6, respectively

$$S|B = \{A_2|B, A_4|B, A_6|B\} = \{2, 4, 6\}$$

$$P[A_2|B] = P[A_4|B] = P[A_6|B] = \frac{1}{3}$$

$$P[S|B] = P[A_2|B \cup A_4|B \cup A_6|B] = P[A_2|B] + P[A_4|B] + P[A_6|B] = 1$$

Law of Multiplication (Product Rule)

Defines the joint probability of occurring two events together

It means – proportion of times two events occur together among all sets of two events:

- If all such pairs are equally likely
- Not independent (1st event affects the 2nd event)

$$P[A|B] = \frac{P[AB]}{P[B]}, P[B] \geq 0$$

$$P[B|A] = \frac{P[AB]}{P[A]}, P[A] \geq 0$$

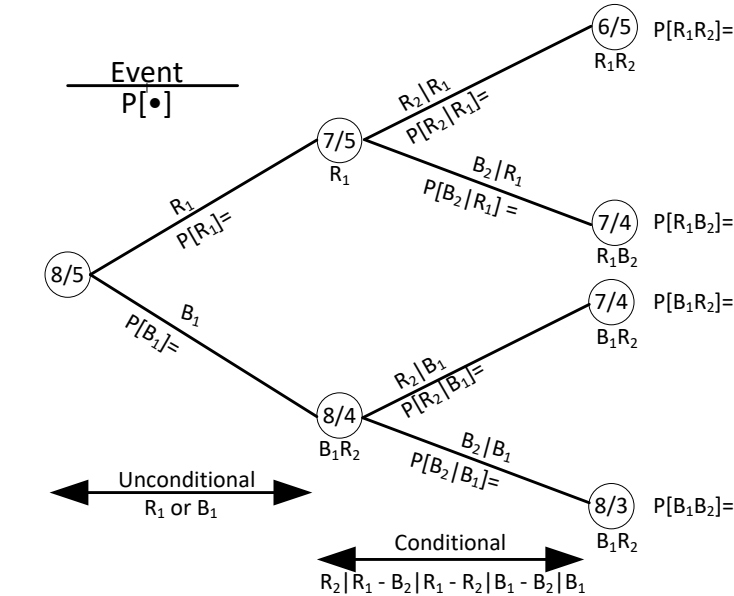
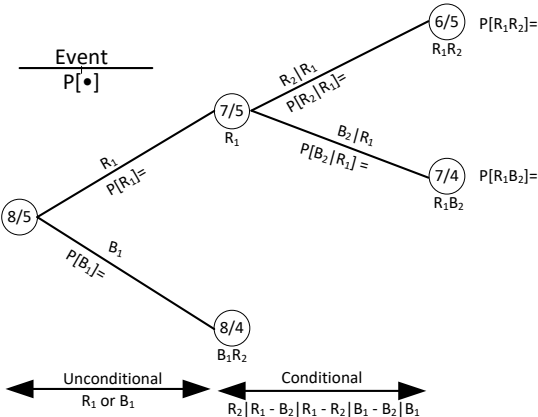
$$P[AB] =$$

Example 1.18: A box contains **8 red** balls and **5 blue** balls. Balls are mixed together. Pick two balls randomly one-after-another without replacement and observe their color. Find $P[R_1R_2]$.

$R_i \triangleq$ Event that the i -th ball is red

$B_i \triangleq$ Event that the i -th ball is blue

$$P[R_1R_2] =$$



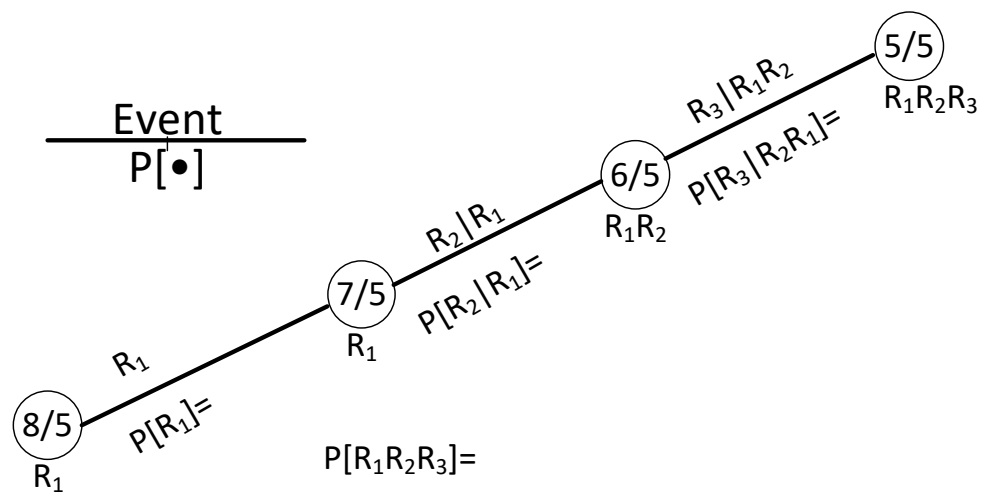
Probability Model

$$S = \{R_1R_2, R_1B_2, B_1R_2, B_1B_2\}$$

The Chain Rule

The joint probability of occurring more than two events can be found by extending the product rule. If A, B, C is a set of 3 events, then

$$P[ABC] =$$



$$P[R_1R_2R_3] =$$

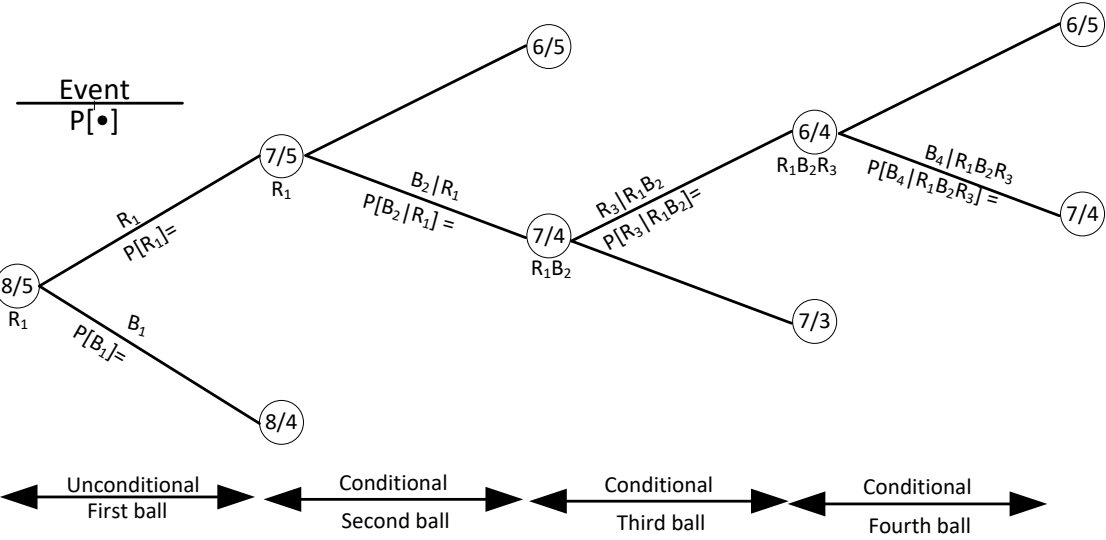
The Chain Rule

The joint probability of occurring more than two events can be found by extending the product rule. If $A_1, A_2, \dots, A_n$ is a set n events, then

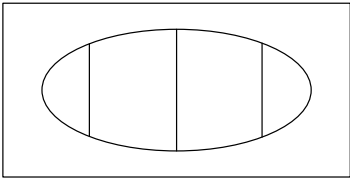
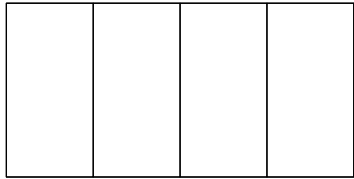
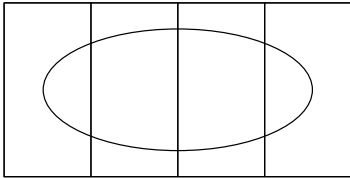
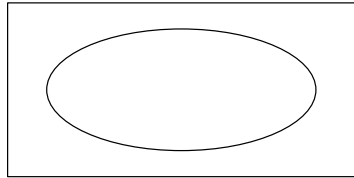
$$P[A_1 A_2 \dots A_n] = P[A_1]P[A_2|A_1]P[A_3|A_1 A_2] \dots P[A_n|A_1 A_2 \dots A_{n-1}]$$

$$= \prod_{i=1}^n P[A_i | \cap_{j=1}^{i-1} A_j]$$

Example 1.19: A box contains 8 red balls and 5 blue balls. Balls are mixed together. Now assume you pick 4 balls successively without replacement. Find the probability that the sequence of colors of the balls is Red, Blue, Red, and Blue.



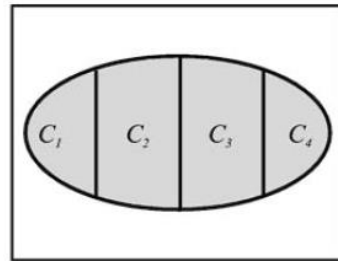
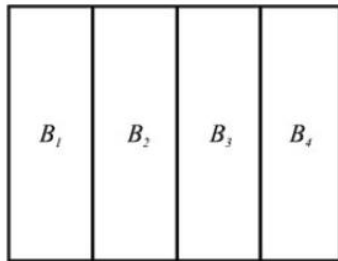
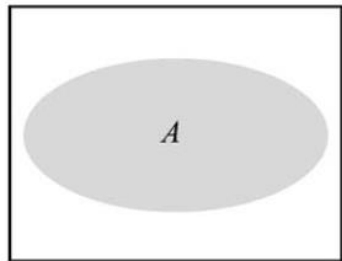
Law of Total Probability



For an event space $B = \{B_1, B_2, \dots, B_n\}$ and any event A in the sample space. Let $C_i = A \cap B_i$, for $i = 1, 2, \dots, n$.

For $i \neq j$, the events C_i and C_j are mutually exclusive and

$$A = C_1 \cup C_2 \cup \dots \cup C_n$$



Law of Total Probability (Sum Rule of Probability):

Let B be an event with $P[B] \geq 0$ and $P[B^c] \geq 0$, then for an event A

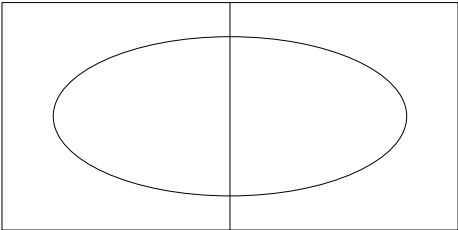
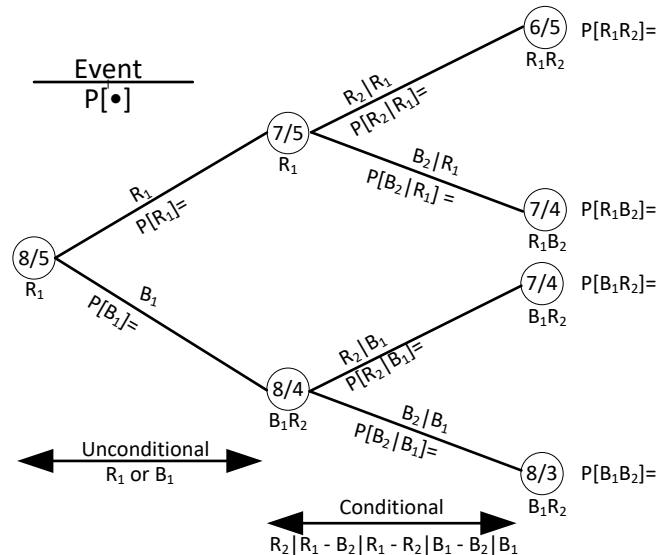
$$P[A] = P[A|B]P[B] + P[A|B^c]P[B^c]$$

Law of Total Probability (Sum Rule of Probability):

Let $B = \{B_1, B_2, \dots, B_n\}$ be an event space, then for an event A

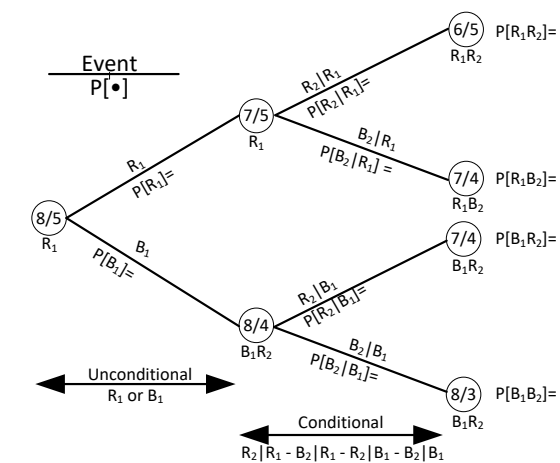
$$\begin{aligned} P[A] &= P[A|B_1]P[B_1] + P[A|B_2]P[B_2] + \dots + P[A|B_n]P[B_n] \\ &= \sum_{i=1}^n P[A|B_i] P[B_i] \end{aligned}$$

Example 1.20: A box contains 8 red balls and 5 blue balls. Balls are mixed together. Pick two balls randomly without replacement and observe their color. Find the probability that the second ball is red.



Bayes Theorem

Example 1.20: A box contains 8 red balls and 5 blue balls. Balls are mixed together. Pick two balls randomly without replacement and observe their color. If the color of the second ball is red, find the probability that the first was also red.



Bayes Theorem

Let $B_1; B_2; \dots, B_n$ be an event space of the sample space S of an experiment. If for $i = 1, 2, \dots, n$, $P[B_i] > 0$ then for any event A of S with $P[A] > 0$,

$$P[B_k|A] = \frac{P[A|B_k]P[B_k]}{P[A|B_1]P[B_1] + P[A|B_2]P[B_2] + \dots + P[A|B_n]P[B_n]}$$

In statistical applications of Bayes' theorem, $B_1, B_2, \dots, B_n$ are called the **hypothesis**, $P(B_i)$ is called the **prior** probability of B_i , and the conditional probability $P[B_i|A]$ is called the **posterior** probability of B_i after the occurrence of A .

Example 1.21 You enter a chess tournament where your probability of winning a game is 0.3 against half the players (call them type 1), 0.4 against a quarter of the players (call them type 2), and 0.5 against the remaining quarter of the players (call them type 3). Suppose you play a game against a randomly chosen opponent, and you win. What is the probability that you had an opponent of type 1?

Independence

Example 1.27: Toss a fair coin twice and observe the sequence of heads and tails

$$S = \{TT, TH, HT, HH\}, \quad P[\{TT\}] = P[\{TH\}] = P[\{HT\}] = P[\{HH\}] = \frac{1}{4}$$

Let

$$A \triangleq \{\text{Head on second toss}\} = \{TH, HH\}$$

$$P[A] = \frac{1}{2}$$

$$B \triangleq \{\text{Tail on first toss}\} = \{TT, TH\}$$

$$P[B] = \frac{1}{2}$$

$$P[A \cap B] = \frac{1}{4}$$

$$P[A|B] = \frac{P[AB]}{P[B]} = \frac{1}{4} \times \frac{2}{1} = \frac{1}{2}$$

$$P[A|B] = P[A]$$

Independence

Knowing that B has occurred, does not change the probability of occurring event A

$$P[A|B] = P[A], P[B] > 0$$

$A \perp B \triangleq A$ is independent of B

$$A \perp B = B \perp A$$

2nd Definition: Independence is defined from the Law of multiplication

$$P[AB] = P[A|B]P[B] = P[A]P[B]$$

$$P[A \cap B] = P[A]P[B]$$

Two events are independent, if their joint probability is the product of their individual probabilities.

Example 1.22 Consider an experiment involving two successive rolls of a 4-sided die in which all 16 possible outcomes are equally likely and have probability $1/16$.

1. Are the events $A_i = \{1\text{st roll results in } i\}$, $B_j = \{2\text{nd roll results in } j\}$, independent?

2. Are the events $A = \{1\text{st roll is a 1}\}$, $B = \{\text{sum of the two rolls is a 5}\}$, independent?

3. Are the events $A = \{\text{maximum of the two rolls is 2}\}$, $B = \{\text{minimum of the two rolls is 2}\}$, independent?

If A is a set of events, events in $A = \{A_1, A_2, \dots, A_n\}$ are independent, if for every subset of $A_i \subset A$

For every subset $A_i \subset A$ with k element, $k = 2, 3, \dots, n$

$$P[A_{i_1}, A_{i_2}, \dots, A_{i_k}] = P[A_{i_1}]P[A_{i_2}] \dots P[A_{i_k}]$$

For example, If the set of events consists of A, B and C then, A, B and C are independent, when

$$P[AB] = P[A]P[B]$$

$$P[AC] = P[A]P[C]$$

$$P[BC] = P[B]P[C]$$

$$P[ABC] = P[A]P[B]P[C]$$

It is possible that the first 3 conditions are satisfied but the last condition is not satisfied and vice versa

Example 1.23: A fair coin is tossed twice and sequence of heads and tails are observed

$$S = \{TT, TH, HT, HH\}$$

$$A = \{\text{head on first toss}\} = \{HT, HH\}$$

$$B = \{\text{head on 2nd toss}\} = \{TH, HH\}$$

$$C = \{\text{1st and 2nd toss produce same outcome}\} = \{TT, HH\}$$

Conditional Probabilities – Further Possibilities

1. Conditional probabilities follow the axioms of probabilities
 - Any event can have the conditional probabilities conditioned on any other events
 - It satisfies all 3 axioms of probability and is a valid probability model

2. Every conditional probability can have further conditional probabilities conditioned on another event
 - Conditional probability of any event A , given another event B $P[A|B]$, can be conditioned on another event C , $P[A|BC]$
 - $P[A|B] = P[A]P[B|A]$
 $P[A|BC] =$

Conditional Product Rule

- $P[AB] = P[A]P[B|A]$ can be conditioned on another event C
 $P[AB|C] = P[A|C]P[B|AC]$
- If $A_1, A_2, \dots, A_n$ and B are events such that $P[B] > 0$ and $P[A_1 \cap A_2 \cap \dots \cap A_{n-1}|B] > 0$, then

$$\begin{aligned} P[A_1 \cap A_2 \cap \dots \cap A_n|B] \\ = P[A_1|B]P[A_2|A_1 \cap B] \dots P[A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1} \cap B] \end{aligned}$$

Conditional Sum Rule: The law of total probability has an analog conditional on another event C , namely,

$$P[A|C] = \sum_{i=1}^n P[A|B_i \cap C] P[B_i|C]$$

Recall,

$$P[A] = \sum_{i=1}^n P[A|B_i]P[B_i]$$

Conditional Independence

Events $A_1, A_2, \dots, A_n$ are conditionally independent given event B , if for every subset $A_{i_1}, A_{i_2}, \dots, A_{i_k}$ of k of these events ($k = 2, 3, \dots, n$),

$$P[A_{i_1} \cap \dots \cap A_{i_k} | B] = P[A_{i_1} | B] \dots P[A_{i_k} | B]$$

Two events A_1 and A_2 are conditionally independent conditioned on B , if

$$P[A_1 \cap A_2 | B] = P[A_1 | B]P[A_2 | B]$$

Which implies (where $P[A_1 B] > 0$),

$$P[A_2 | A_1 B] = P[A_2 | B]$$

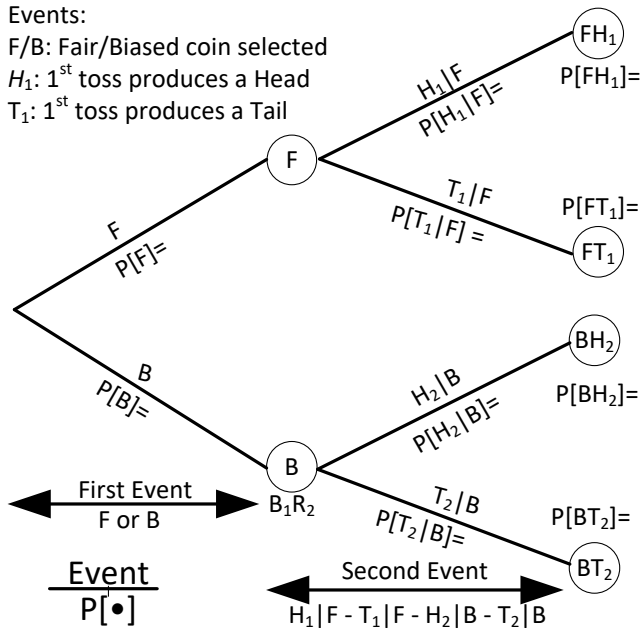
Conditional Bayes' Theorem

$$P[B_i|A \cap C] = \frac{P[A|B_i \cap C]P[B_i|C]}{\sum_{j=1}^n P[A|B_j \cap C]P[B_j|C]}$$

Recall the general Bayes' Theorem

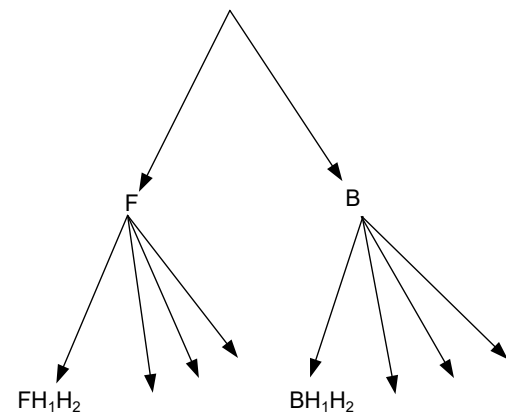
$$P[B_i|A] = \frac{P[A|B_i]P[B_i]}{\sum_{j=1}^n P[A|B_j]P[B_j]}$$

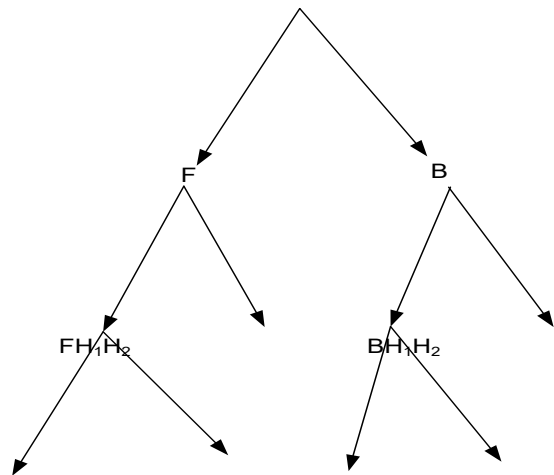
Example 1.24: Suppose a box contains one fair and one double headed coin. Suppose a coin is selected randomly and a head is obtained. Find the probability that the selected coin is the fair coin.



Example 1.24 (Continued): Suppose the same coin is tossed again and that the two tosses are conditionally independent given both F and B . Suppose further that another head is obtained. Find the probability that the selected coin is fair.

$$P[F|H_1H_2] =$$





$$\Pr(B_1|H_1 \cap H_2) = \frac{\Pr(B_1|H_1) \Pr(H_2|B_1 \cap H_1)}{\Pr(B_1|H_1) \Pr(H_2|B_1 \cap H_1) + \Pr(B_2|H_1) \Pr(H_2|B_2 \cap H_1)}$$

$$= \frac{(1/3)(1/2)}{(1/3)(1/2) + (2/3)(1)} = \frac{1}{5}.$$

